



ADAS+DMS System

QD003

Data sheet



1. PRODUCT OVERVIEW

The safe driving assistance terminal QD003 is an on-board terminal device in the informationalized system for vehicle comprehensive monitoring.

It uses AI algorithms to process data from active safety cameras. Advanced Driver Assistance Systems (ADAS) can notify the driver when a lane change occurs accidentally or before a crash. Driver Monitoring System (DMS) can analyze the driver's fatigue and distraction in real time and issue timely warnings. Blind Spot Detection (BSD) can detect the blind area according to the distance between the blind area and the vehicle body. If there is a target within its detection range, it will sound an alert to prevent potentially dangerous accidents.

2. PRODUCT APPEARANCE AND SPECIFICATIONS

2.1 Product Appearance



MDVR



DMS Camera



ADAS Camera



BSD Camera (optional)



ADAS Alarm (optional)

2.2 MDVR Specification

Items	Specification
Chip	SSC 8838G
System	linux
Storage	1GB+8GB
Video Inputs	Maximum 4; 1080P; AHDAccess
Audio	4 video ports; support audio input
Video Output	CVBS
Communications	LTE CAT 4
Position	GNSS; RTK (optional)
Network Port	100M (optional)
External Ports	RS232, RS485, CAN×2
I/O	Right & Left Turns; Vehicle Reversing; One-touch Emergency Button; Open Door With High Beams, Close Door With Low Beams; Fuel Cut; Power Cut
Horn	Two channels: one external (optional) and one internal
Voice Announcement	TTS
Function	ADAS, DMS, two-channel BSD, HOD
Protocol Standards	JT/T 808-2019, JT/T 794-2019 , JT/T 1076-2016, JT/T 1078-2016
Rated Voltage	9 ~ 36V DC

Rated Power	≤10W
Operating Temperature	-20 ~ 70℃
Operating Humidity	10 ~ 90%RH
Size	160mm×105mm×40mm (W*D*H)
Material	PC Plastic, aluminum alloy, electrolytic plate
Protection Level	IP54

2.3 DMS Camera Specification

Items	Specification
Vision Sensors	1/3", CMOS
Effective Pixels	1280×720
Field of View	58° (Horizontal)
Focus	40~200cm
Transmission Method	AHD

2.4 ADAS Camera Specification

Items	Specification
Vision Sensors	1/2.9", CMOS
Effective Pixels	1280×720
Field of View	51° (Horizontal)
Focus	200cm~∞
Transmission Method	AHD

2.5 BSD (Optional) Camera Specification

Items	Specification
Resolution	1080P
Field Of View	≈90°
Focus	2m~∞
iR Compensation	Wavelength: 850nm; Lighting range: 15-20 meters
Protection Level	IP67
Transmission Method	IPC

3. PRODUCT FUNCTION, SIZE AND PORTS DESCRIPTION

3.1 Product Function (DMS Camera)

Warning Type	Description	Remark
Drowsy Driving Warning	Identify continuous eye closure and yawning	Standard
Distraction Warning	Identify the continuous turning of the head left and right and the state of lowering/raising the head	Standard
Make/Receive Phone Calls Warning	Identify the gesture of phone making/receiving	Standard
Smoking Warning	Identify the gesture of smoking	Standard
Off-Duty Warning	Identify when the driver leaves the seat	Standard
Driver Change Notice	The current detected face is different from the last one	Standard

Long Drives Warning	Continuous 4 hours of driving/8 hours of driving in a 24-hour period	Optional
Not Wearing Seat Belt Warning	Identify the status of the seat belt	Optional
Drinking Water Warning	Identify the gesture of drinking water	Optional
Hands Off The Steering Wheel Warning	Identify whether the driver's hands are holding the steering wheel	Optional (HOD Camera required)
Phone Usage Warning	Identify mobile phone and human arm posture	Optional (HOD Camera required)
Failure Warning	Camera blocked warning	Standard
	Infrared glasses warning	Standard

Note:

- 1) The driver can be recognized normally during the day and at night, whether wearing myopia glasses or ordinary sunglasses.
- 2) The enabling speed can be configured according to actual usage.
- 3) DMS is highly resistant to external light sources, and can work normally under different lighting scenarios, such as daytime, cloudy days or nighttime.

4) For more functional requirements, please contact official technical support.

3.2 Product Function (ADAS Camera)

Warning Type	Description	Default Enabling Conditions	Remark
Forward Collision Warning (FCW)	Monitor in real time the distance to the vehicle ahead. The driver can be warned as early as 2.7 seconds before a collision is expected to occur, thus preventing rear-end collisions.	Speed $\geq$ 30km/h	Standard
Lane Departure Warning (LDW)	Monitor the current lane markings and vehicle trajectory. When the vehicle is about to accidentally deviate from the lane, a warning will be issued to remind the driver to stay in a safe lane.	Speed $\geq$ 55km/h	Standard
Headway Monitoring and Warning (HMW)	Display the distance to the vehicle ahead (in seconds) and warn the driver when the vehicle approaches a preset danger distance.	Speed $\geq$ 30km/h	Standard
Urban Forward Collision Warning (UFCW)	Monitor the distance to the vehicle ahead when driving at low speeds, and remind the driver to avoid collision when the actual distance is less than the safe distance.	Speed < 30km/h	Optional
Pedestrian Collision Warning (PCW)	Identify pedestrians ahead and detect distance; issue an alarm in dangerous situations to remind the driver to keep a safe distance from pedestrians.	Speed < 50km/h	Optional

Speed Limit Indicator (SLI)	Identify speed limit signs and their speed limit values, and issue an alarm when the vehicle exceeds the speed limit.	/	Optional
Stop & Go	Monitor the vehicle in front when parking at a close distance (such as waiting in line at a red light), and remind the driver to start the vehicle in time when the vehicle in front starts moving forward from a standstill, so as to ensure orderly driving.	/	Optional
Rapid Acceleration Warning (RAW)/ Rapid Deceleration Warning (RDW)	When the vehicle accelerates or brakes suddenly, the driver is reminded to control the speed reasonably to avoid safety risks caused by emergency speed changes.	/	Optional
Sharp Turn Warning (STW)	When the vehicle changes lanes or reaches an extreme state while driving on a curve, the driver is reminded to slow down or reduce the turning direction immediately to avoid an accident.	/	Optional

Note:

- 1) The enabling speed can be configured according to actual usage;
- 2) It can be used in most scenarios, such as daytime, nighttime, tunnels, rainy days, etc. (except extreme weather);
- 3) Support ADAS camera blocked warning;
- 4) For more functional requirements, please contact official

technical support.

3.3 Product Function (BSD Camera; Optional)

Item	Description
Right blind spot monitoring	The right blind spot monitoring function uses a high-security camera at the rear of the vehicle to detect pedestrians on the right side. If a pedestrian is too close to the vehicle, an alarm will be triggered to remind the driver; at the same time, the sound and light alarm outside the vehicle will remind pedestrians to stay away from the vehicle. When the identified target is detected to enter the alarm area and the alarm condition threshold is reached, a voice or visual display alarm prompt will be issued.

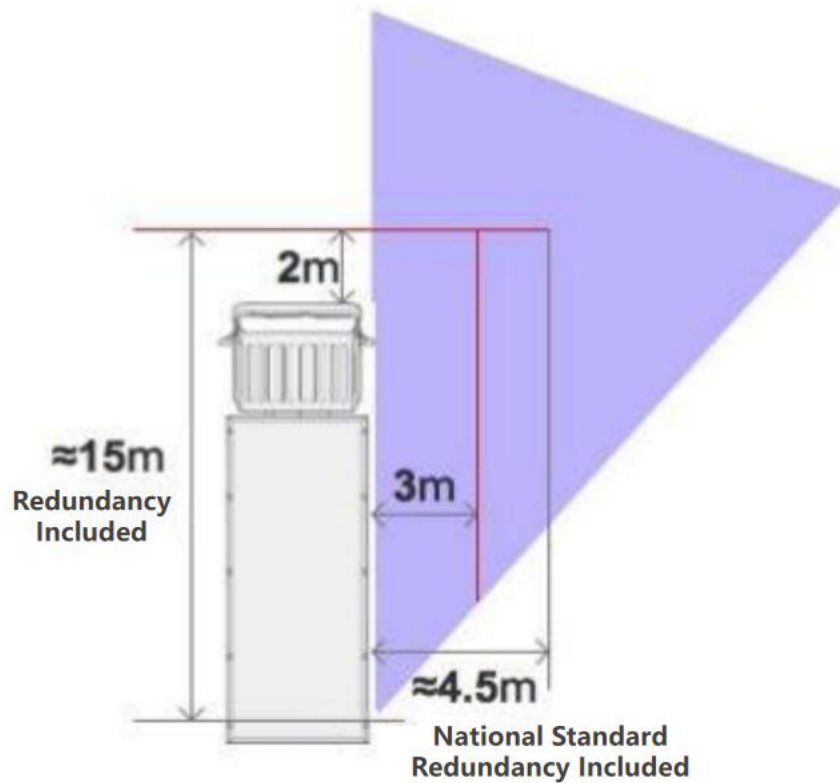
Working speed: 0-30km/h

There are three levels of alarm, based on the lateral distance between the target and the vehicle (L):

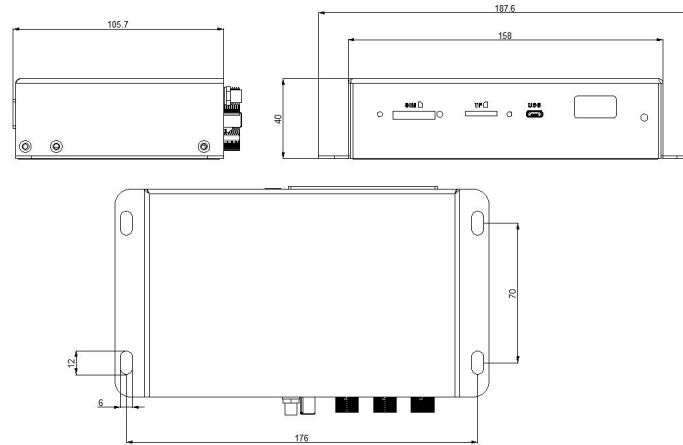
Level 1 alarm: $L \leq 1\text{m}$.

Level 2 alarm: $1\text{m} < L \leq 2\text{m}$.

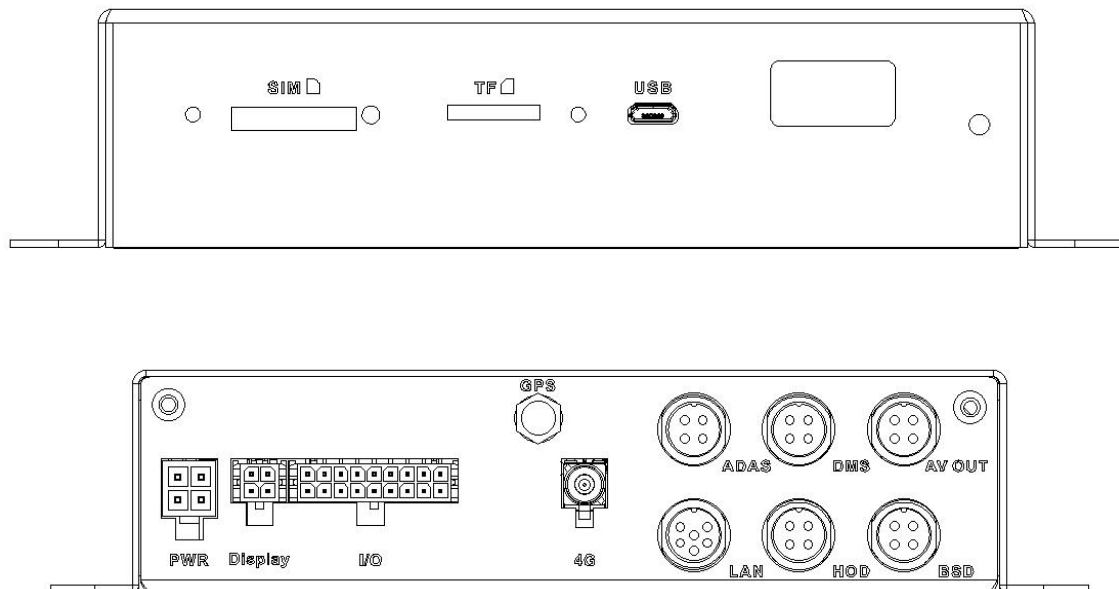
Level 3 alarm: $2\text{m} < L \leq 3\text{m}$.



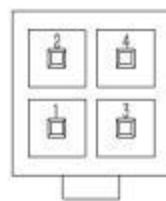
Warning Prompts	Actions	Remark
Indoor	The MDVR speaker issues a voice alarm: Level 1: Danger Level 2: Please pay attention to pedestrians Level 3: Pedestrians on the right	Warning level: Level 1 > Level 2 > Level 3
Outdoor	When Level 1 and Level 2 alarms are triggered, the sound and light alarms flash and a voice alarm sounds: Danger! Please stay away from the vehicle (lasting for 4s)	The sound and light alarm does not work in Level 3 alarm.



3. 6 Description of Ports on the Front and Rear Panels



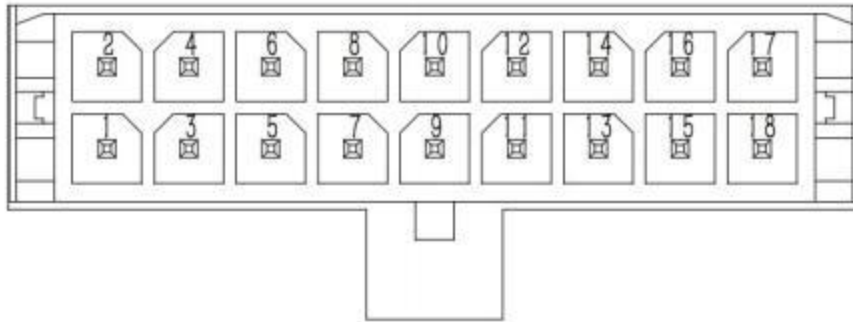
3.7 Specifications of Power Ports



Pins	Function	Remark
1	Input positive	
2	Earth	

3	Earth	
4	ACC	

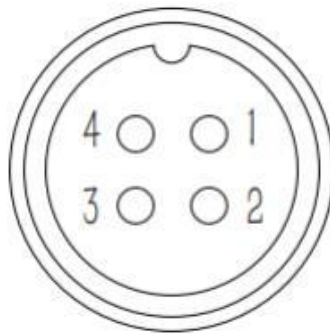
3.8 Specifications of I/O Ports



Pins	Function	Description
1	DC5V	Power output pin.
2	Earth	
3	Turn right	
4	RS232_TX	
5	Simulated Vehicle Speed	
6	RS232_RX	
7	Turn Left	
8	I/O Input	Reverse signal input.
9	CAN_H	
10	I/OOutput	Connect external alarms, such as seat and steering wheel vibrations, oil cut and power cut.
11	Vehicle CAN_L	
12	I/O Input	Low beam signal input or door closing detection
13	I/O Input	High beam signal input or door opening detection

14	I/O Input	Emergency alarm input.
15	RS485_A	
16	Speaker+	Speaker for IP calls
17	RS485_B	
18	Trumpet-	Speaker for IP calls

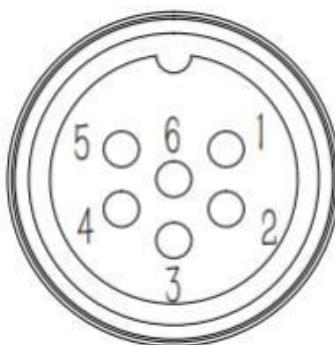
3.9 Definition of AHD Ports



Pins	Function
1	Video Output 12V Power +
2	Video Output 12V Power -
3	Audio
4	Video Signal Input

3.10 Specifications of LAN ports

The LAN Ethernet port specifications are as follows:

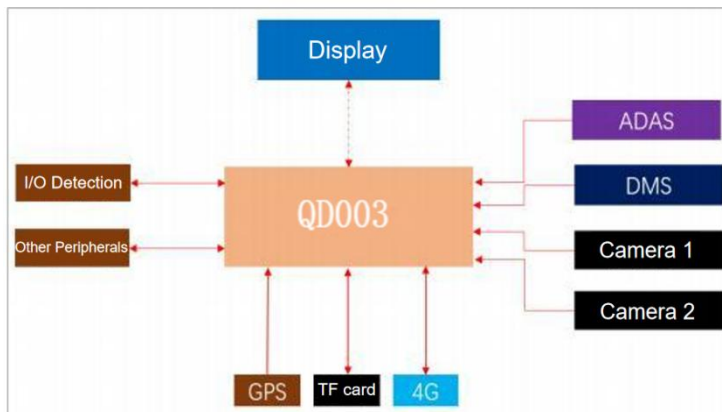


Pin number	Signal
1	TX -

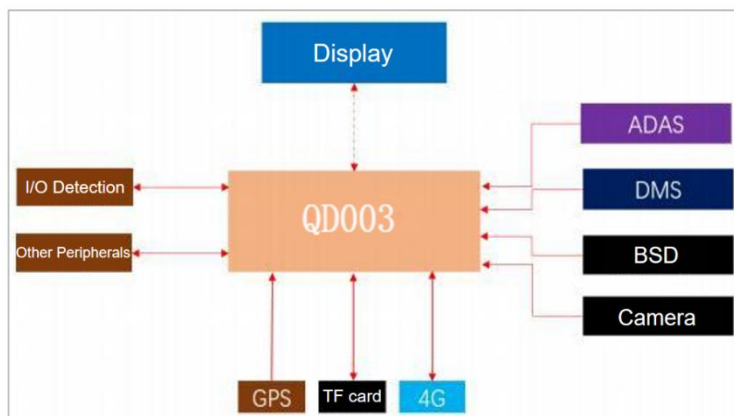
2	TX+
3	+12V
4	RX -
5	RX+
6	GND

4 . SOLUTION PORTFOLIO

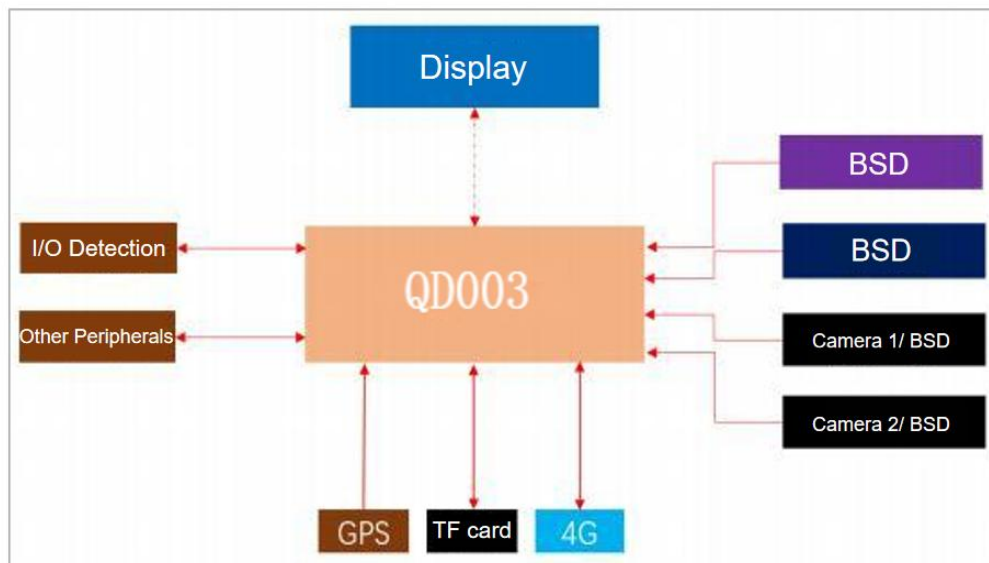
4.1 Active Safety Package



4.2 Monitoring/Active Safety + BSD



4.3 Monitoring + Multi-channel BSD



4.4 Monitoring/Active Safety + Cargo Monitoring Camera: N1

